DTU

46770 Integrated Energy Grids

Aleksander Grochowicz

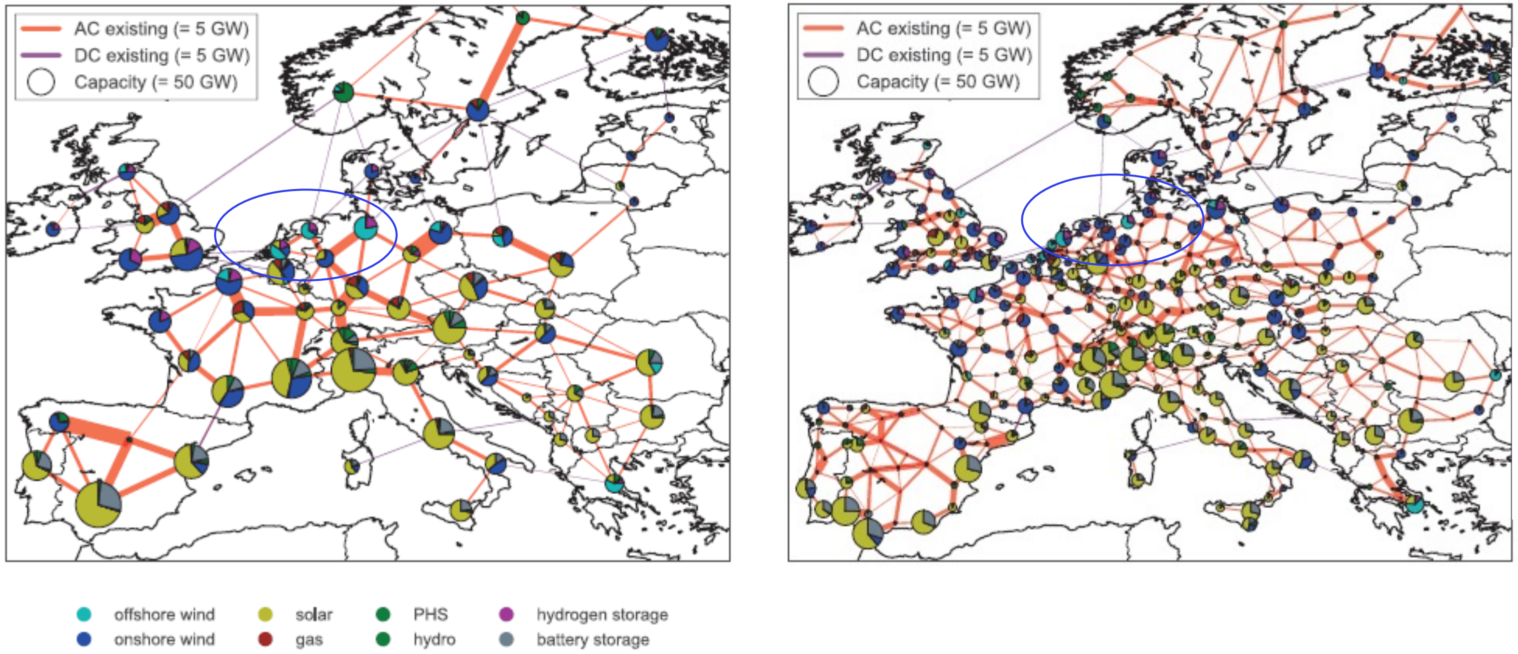
Lecture 12 – Multicarrier energy systems II (industry, aviation, shipping)

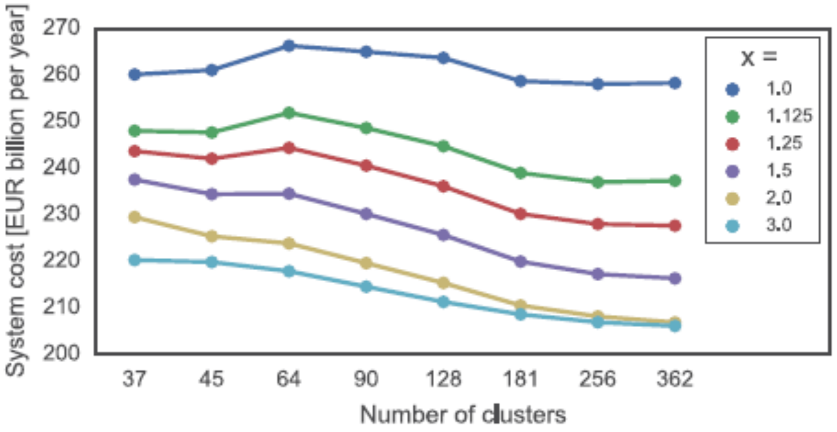
Recap: Which spatial and temporal resolution is good enough?

Which spatial resolution is good enough?

When increasing the number of nodes, the system cost remains roughly constant due to the counterbalancing of two effects:
(a) sites with high capacity factors for wind and solar are available for a more finely resolved network,

(b) but the emergence of bottlenecks inside countries prevents the use wind energy generated at exterior nodes



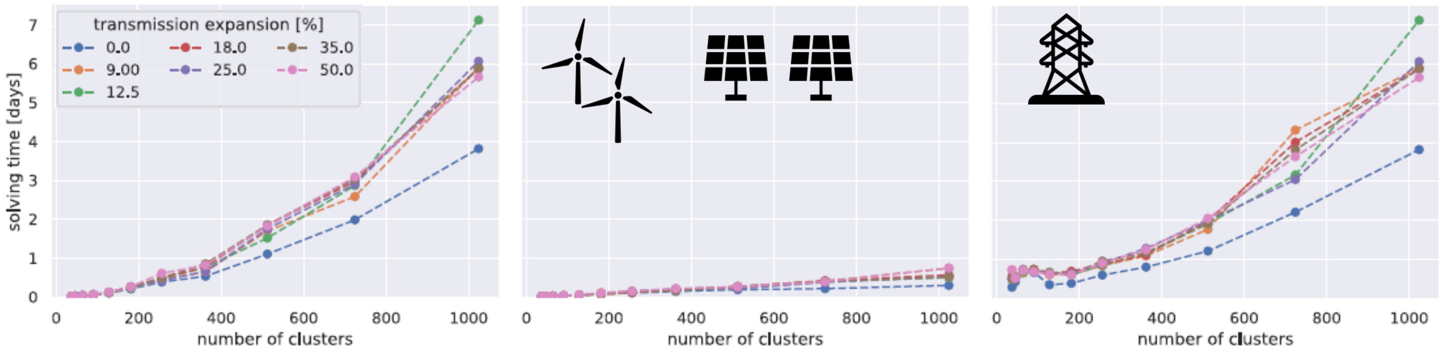


Hörsch and Brown, IEEE (2017)

Higher spatial resolution requires more computation capabilities and downscaling historical information that is typically provided at country level.

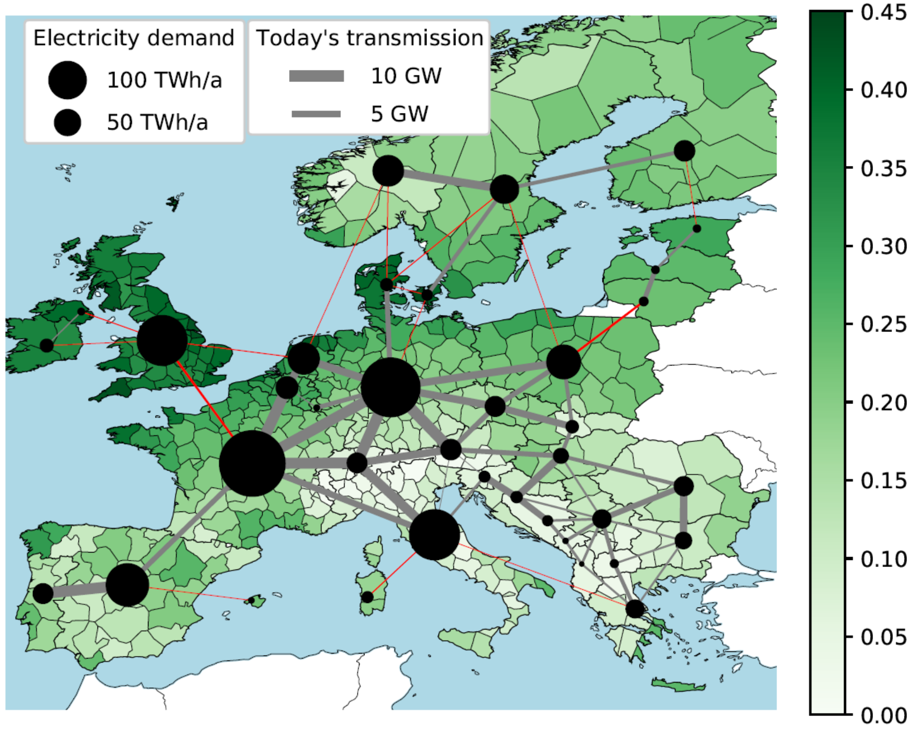
Which spatial resolution is good enough?

Attaining high resolution on transmission nodes increases solving time, but increasing (only) resolution on sites for renewable has negligible impact.



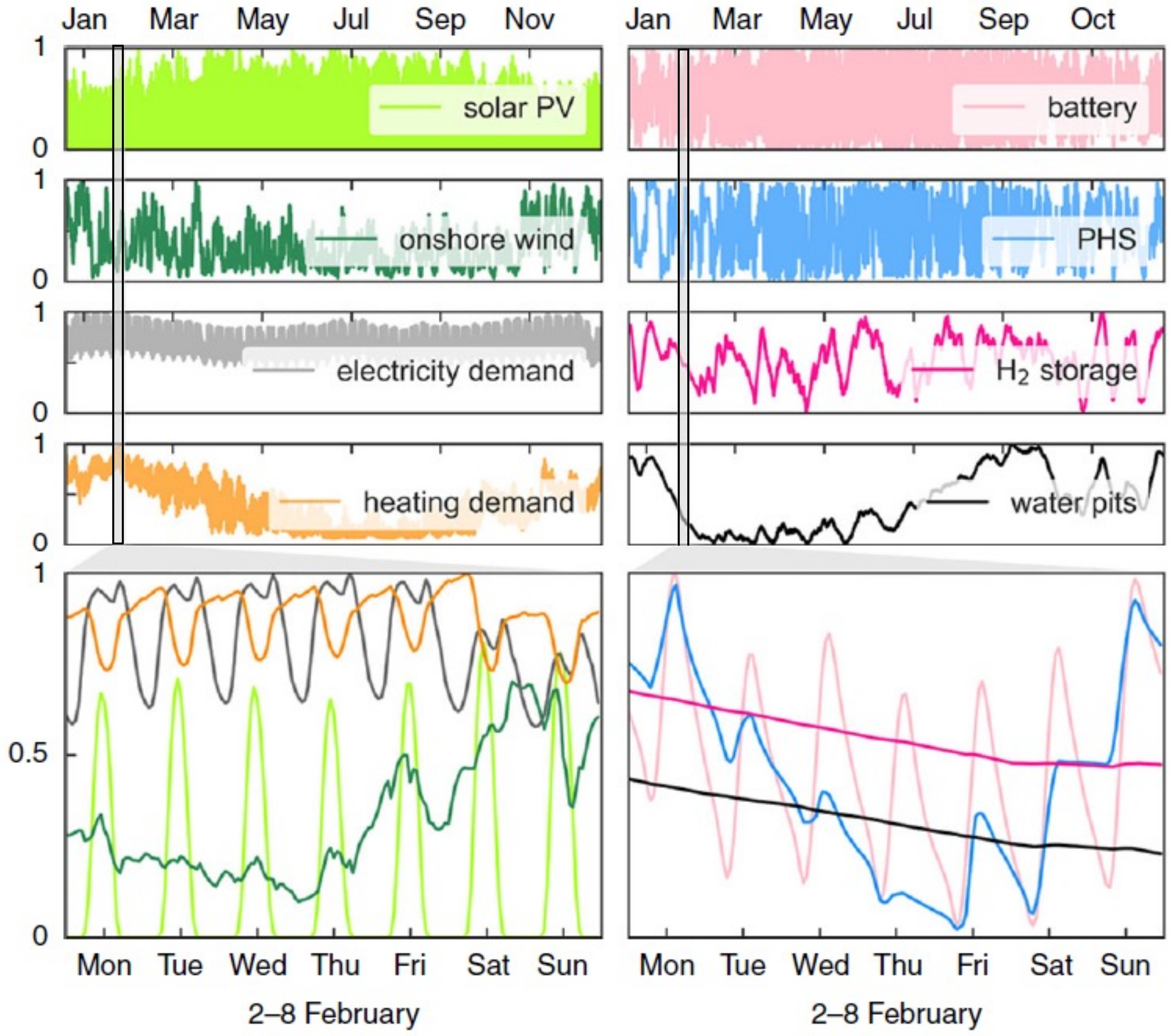
Frysztacki et al., Applied Energy (2021)

In our recent works, we keep a 37-node network and 370 nodes to resolve solar and wind resource.



Victoria et al., Joule (2022)

Which time resolution is good enough?



Victoria et al., Nature Commun. (2020)

We need **uninterrupted hourly time resolution for a full year** to capture the main fluctuations:

- solar and wind power generation smoothed by the grid and storage
- the role of long-term storage
- system operation during dark doldrums/dunkelflauten (i.e. periods with low wind and solar generation)

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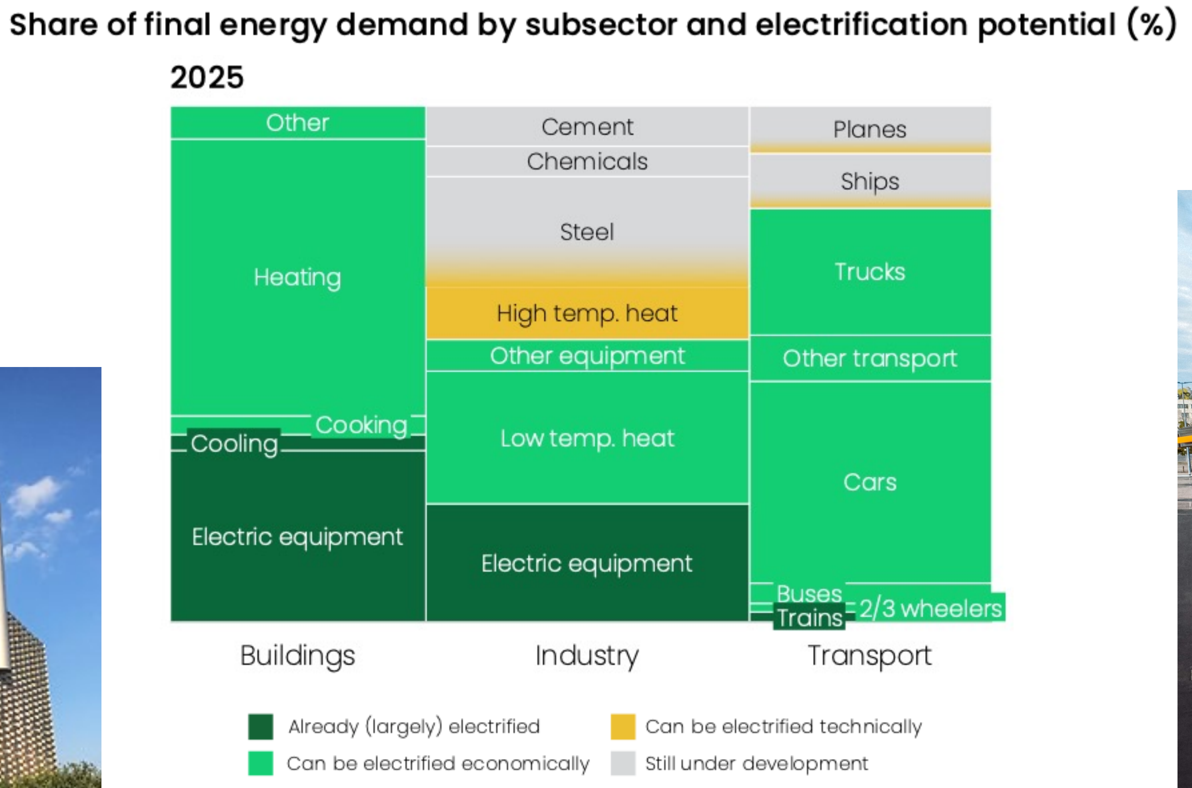
Lecture 12 – Multicarrier energy systems II (industry, aviation, shipping)

What if we can’t electrify a sector?

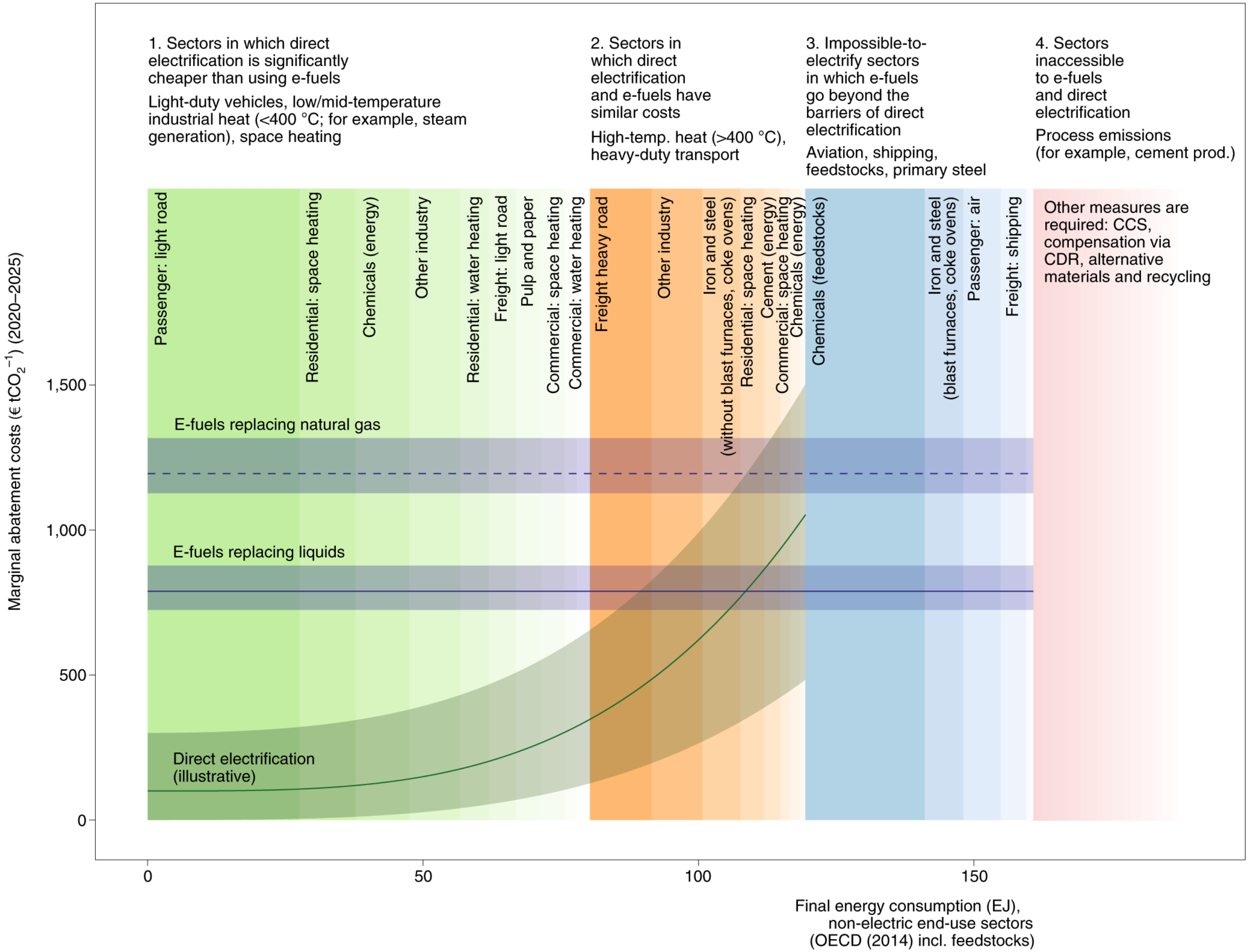


Source: Ember: the electrotech revolution

Learning goals

- Present the main challenges and opportunities introduced by muti-carrier energy systems comprising power-to-X technologies.
- Describe modelling approaches that can be used to represent shipping and aviation demand and strategies to decarbonise those sectors.
- Describe modelling approaches that can be used to represent industry demand and technologies to decarbonise the different industrial sectors.
- Describe modelling approaches that can be used to represent power-to-X technologies.
- Formulate optimisation problems comprising power, shipping, aviation and industry sectors in the computer.

Some sectors are impossible to electrify, and some emissions cannot be avoided



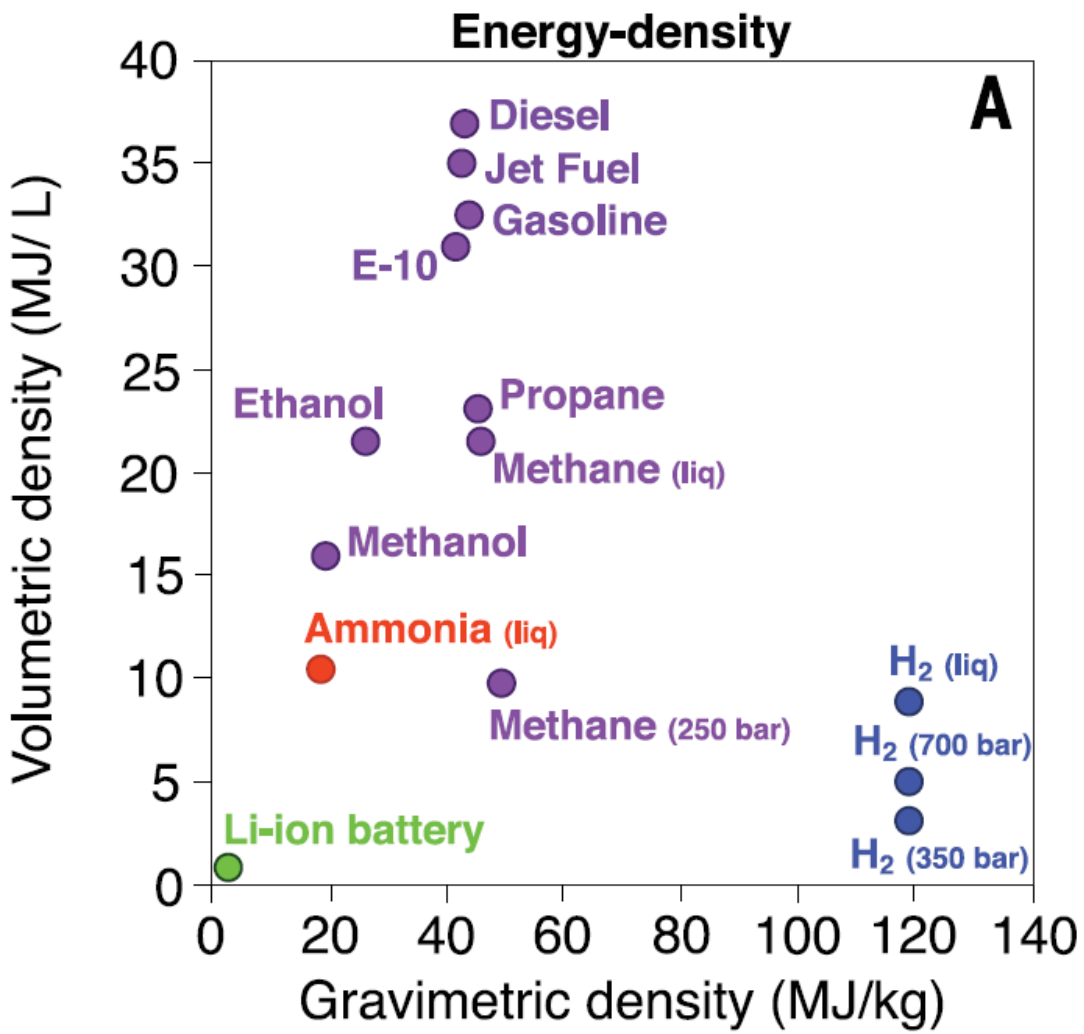
Ref: Ueckerdt et al, 2021, Nature Climate Change

Modelling shipping/aviation demand and flexibility

Shipping and aviation

Challenge: shipping and aviation require fuels with good volumetric and gravimetric density

Opportunity: synthetic fuels (hydrogen, methanol, methane, oil) can be produced when there is an excess of renewables providing flexibility



Ref: Davis et al, Science (2018)

Shipping

Ammonia can be produced via Haber-Bosch process

$N_2 + 3\ H_2 \rightarrow 2\ NH_3$

Methanol can be produced in methanolisation units

$CO_2 + 3\ H_2 \rightarrow CH_3OH + H_2O$

Shipping

	Hydrogen (H_2)	Ammonia (NH_3)	Methanol (CH_3OH)
Fuel properties under ambient conditions	gas	gas	liquid
Fuel energy density	8.5 MJ/L for liquid 120 MJ/kg	11.6 MJ/L 22.5 MJ/kg	15.7 MJ/L 19.5 MJ/kg
Engine TRL*	4	6	9
Toxicity	Non-toxic	Very high	Lower than oil
Key benefit	No efficiency losses for conversion into ammonia or methanol	Does not contain carbon (no need to previously capture CO_2)	Easy to convert a ship to use methanol (allow dual-fuel engine)

* TRL: technology readiness level (1-9)

Aviation

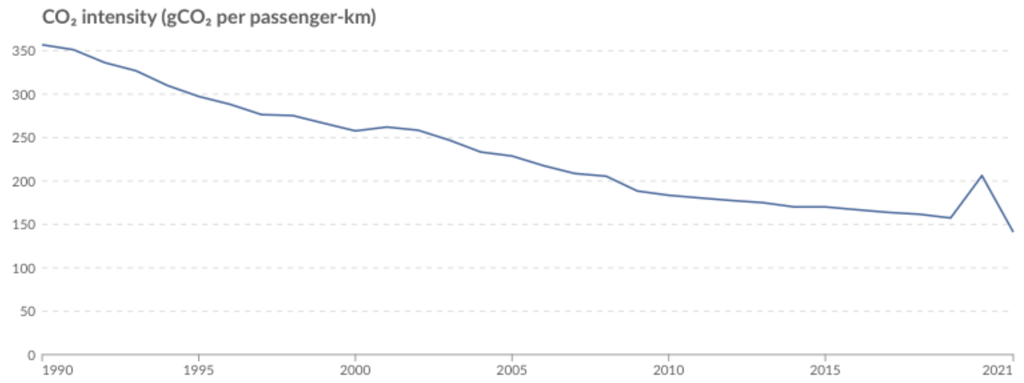
Aviation is responsible for 2.5% of global CO_2 emissions (increasing share), but accounts for 4% of global warming (2/3 coming from non-CO2 forcings) to date.

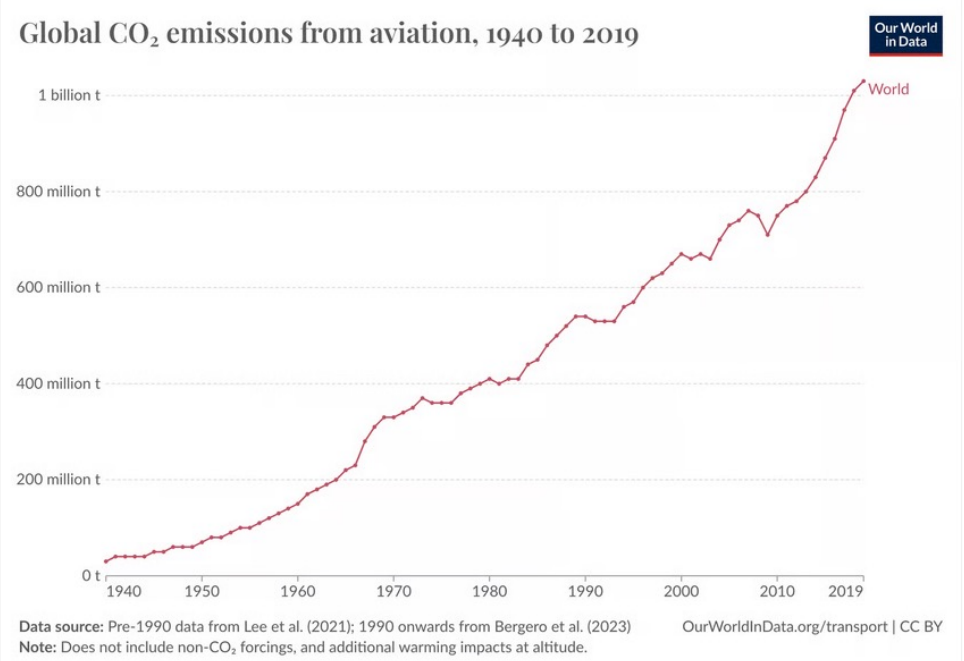
Synthetic hydrocarbon can be produced via Fischer-Tropsch process

$(2n + 1)\, H_2 + n\, CO \rightarrow C_nH_{2n+2} + n\, H_2O$

Energy expensive to produce.

Alternative strategies being discussed for this sector: biomass-based jet fuel, demand reduction, offset by negative emissions.

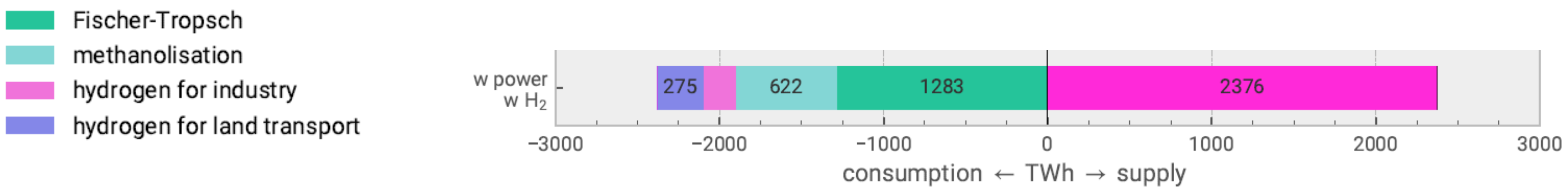




Hydrogen demand, production, storge and transport

Hydrogen demand

Hydrogen is mostly needed to produce synthetic fuels:



Neumann, Zeyen, Victoria, Brown, Joule, 2023

Up-to-date information: Johnson et al., 2025 Realistic roles for hydrogen in the future energy transition

Hydrogen production

Current situation:

H_2 produced from methane

$CH_4 + H_2O \text{ -> } CO + 3H_2$

$CO + H_2O \text{ -> } CO_2 + H_2$

Possible alternative routes in the future:

Blue hydrogen: produced from methane while capturing the CO_2

(current capture rate around 90%)

Green hydrogen: produced using renewable electricity

$2\ H_2O \text{ -> } 2\ H_2 + O_2$

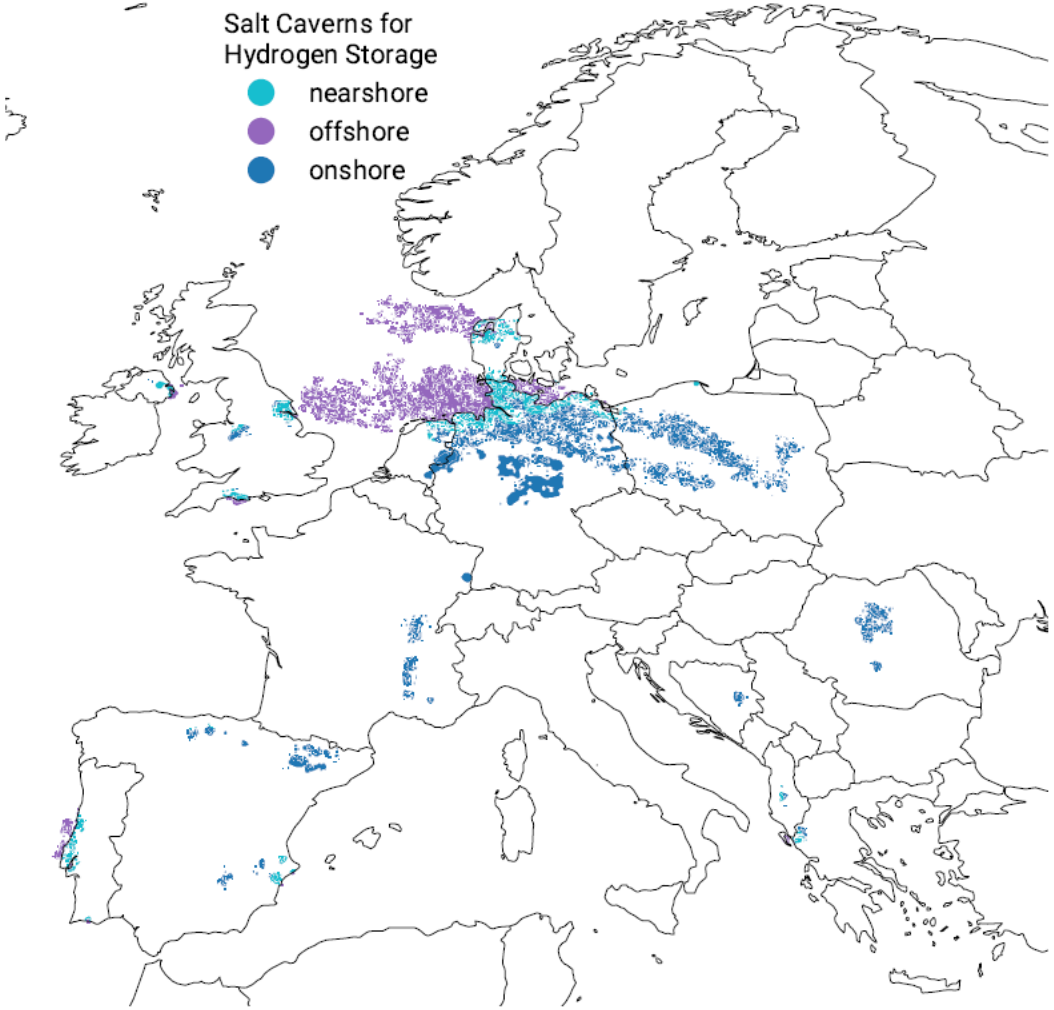
Pink hydrogen: produced using nuclear electricity

Which one has lower cost? Depends on electrolyser cost, CO_2 and electricity prices, and CO_2 capture rates.

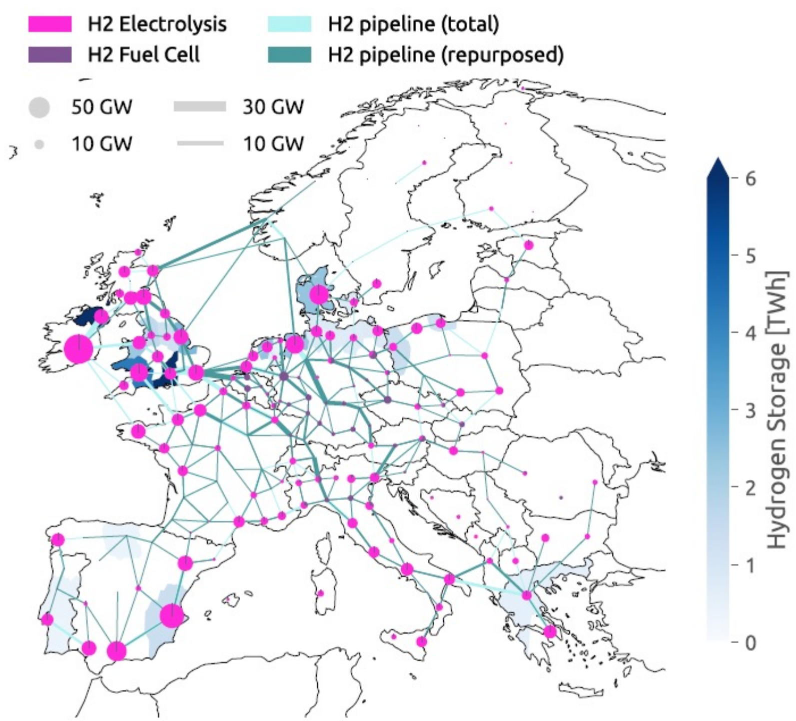
Hydrogen storage

Hydrogen can be easily stored in overground tanks

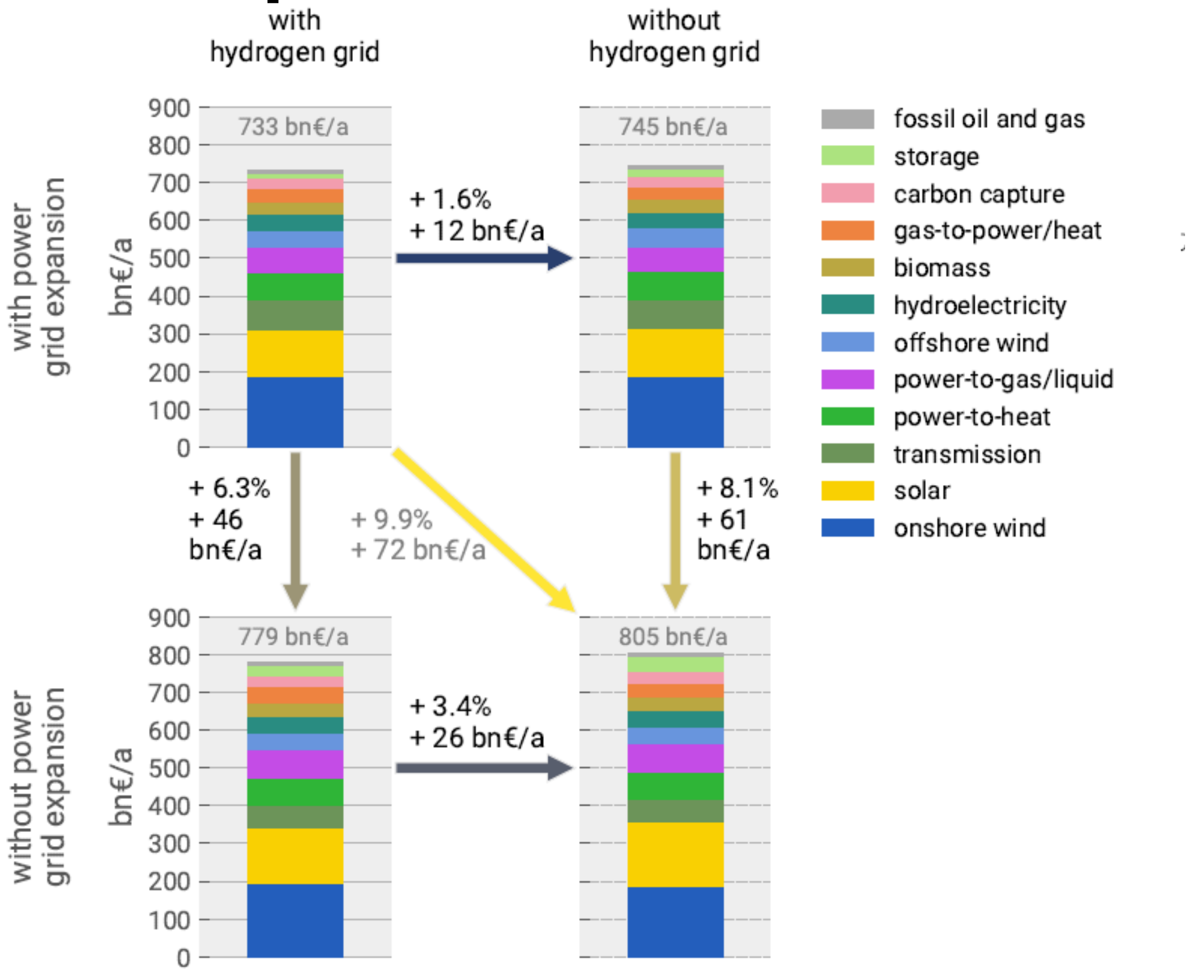
Hydrogen can also be stored in underground salt caverns



Is a H_2 grid cost-effective for Europe under net-zero?



Neuman, Zeyen, Victoria, Brown, Joule, 2023

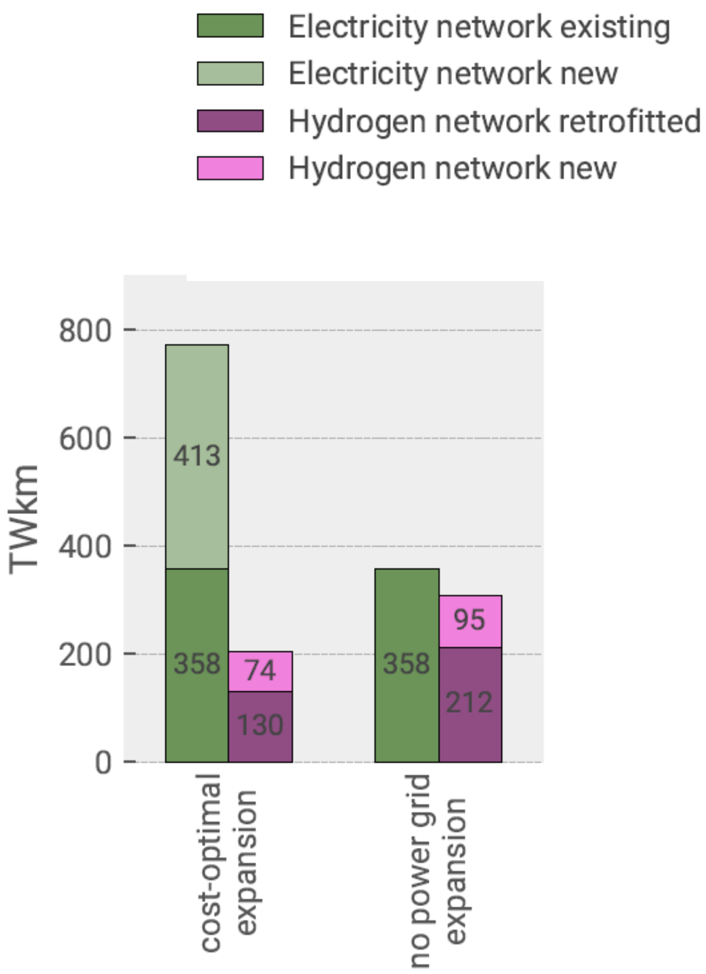


H_2 and electricity transmission grid play complementary roles but:

- Not building H_2 network increases system cost by 1.6%
- Not expanding electricity transmission grid increases cost by 6.3%

Is a H_2 grid cost-effective for Europe under net-zero?

Electricity transmission grid doubles its capacity, H_2 network mostly repurposed from gas network

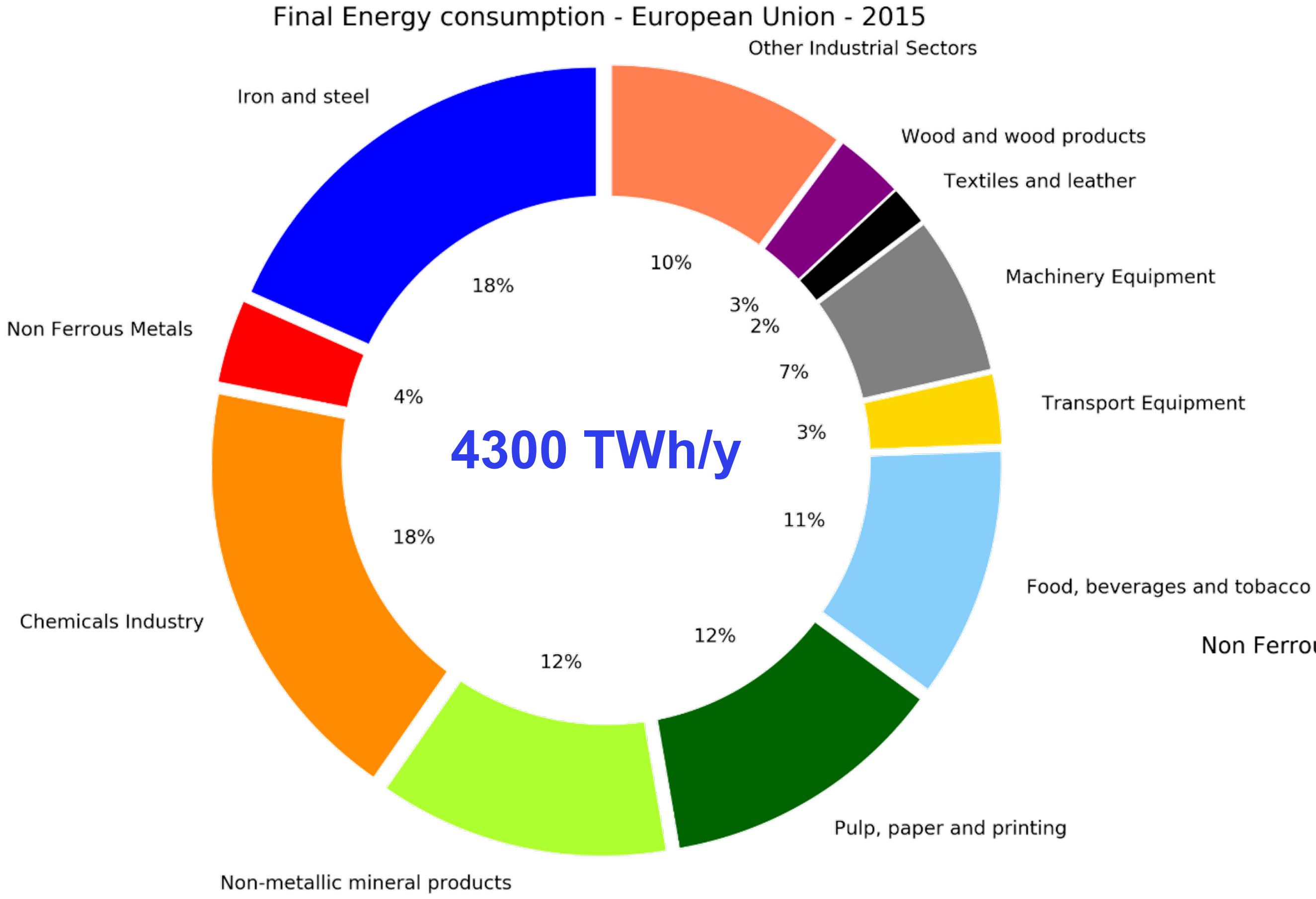


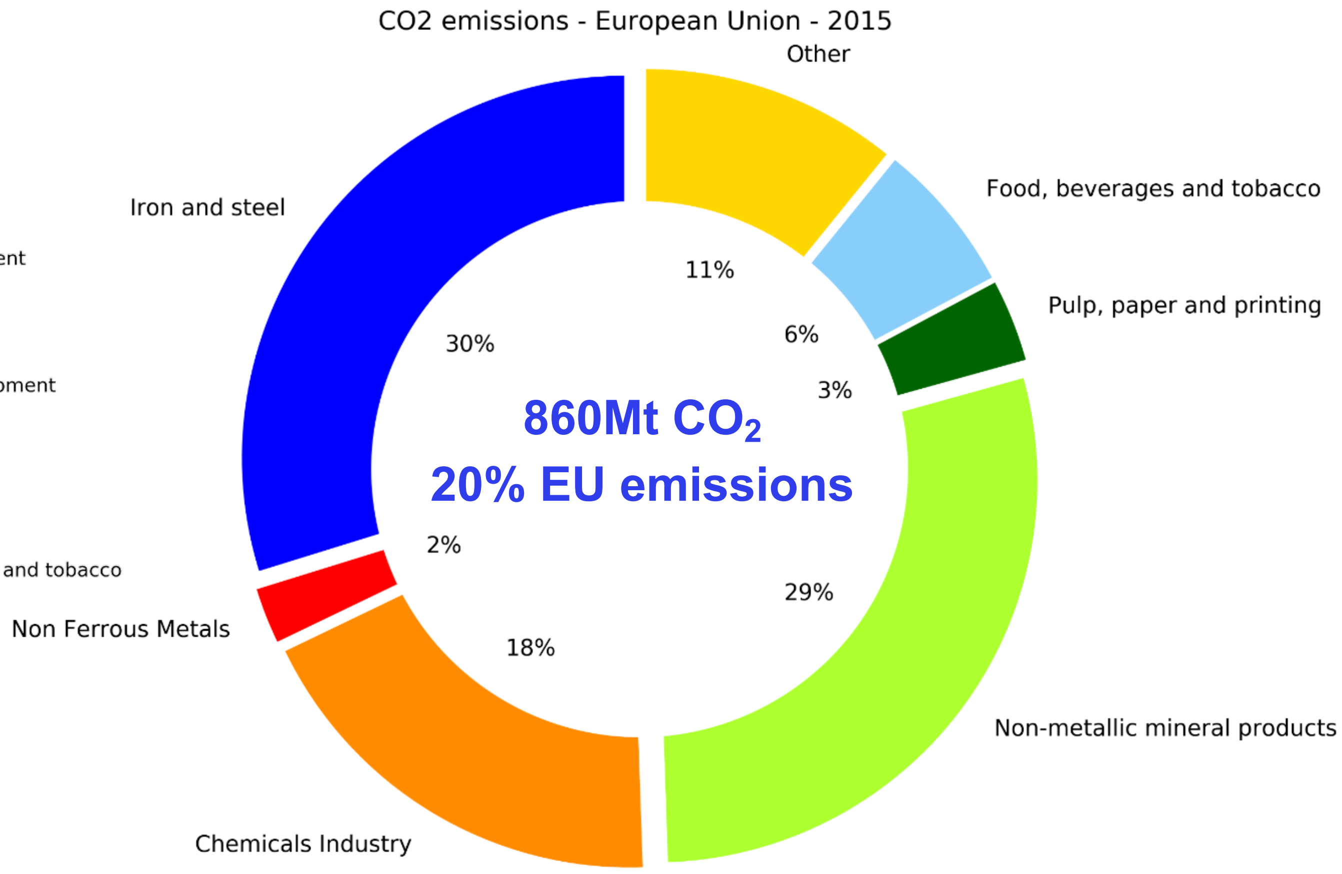
Neuman, Zeyen, Victoria, Brown, Joule, 2023

Modelling industry demand and flexibility

Current status

There are different industry sectors with different characteristics





L. Mantzos, *et al*., JRC-IDEES: Integrated Database of the European Energy Sectors

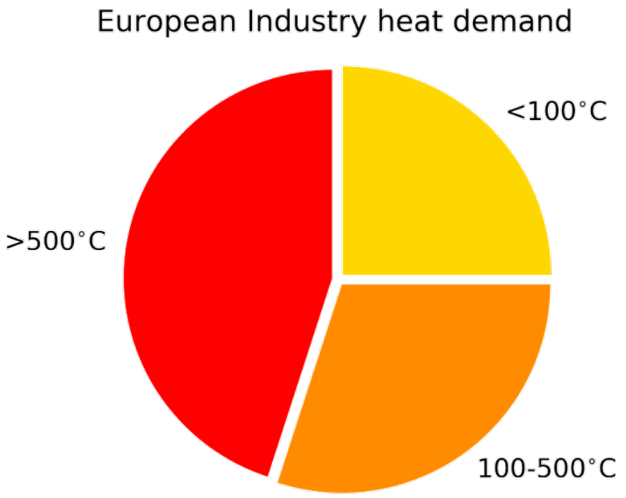
Industry demand

Challenges: supply of high-temperature heat and process emissions

Industry CO_2 emissions = Energy supply + Process emissions

Strategies to reduce emissions

Use low-carbon energy sources



M. Rehfeldt, et al. Energy Efficiency 11, 2018

Alternative manufacturing processes

Reduce required virgin materials by recycling and circular economy practices

Opportunities: flexible electricity demand, synthetic fuels (hydrogen, methanol, methane, oil) can be produced when there is an excess of renewables providing flexibility

Iron and Steel

Current situation:

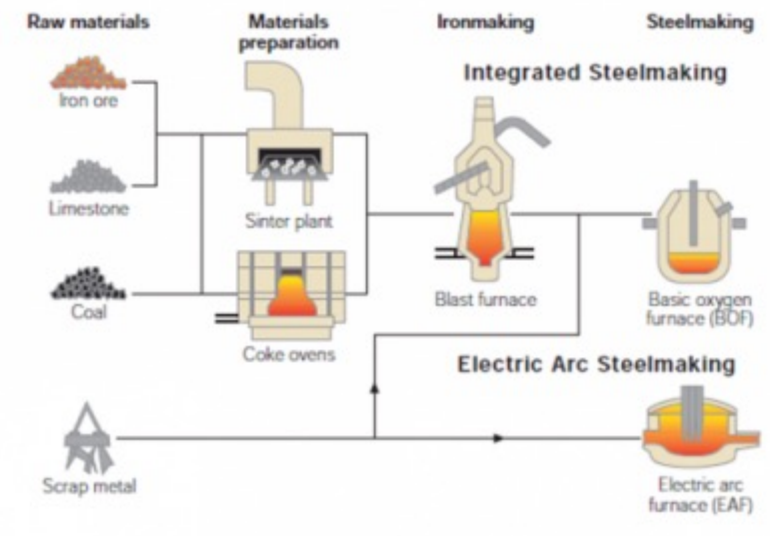
- 60% Primary route, integrated steelworks :

Fe_2O_3+3CO → 2Fe +3CO_2 0.22 tCO_2/t steel

Process emissions !

FeO+CO → Fe +CO_2

FeO+CO → Fe +CO_2

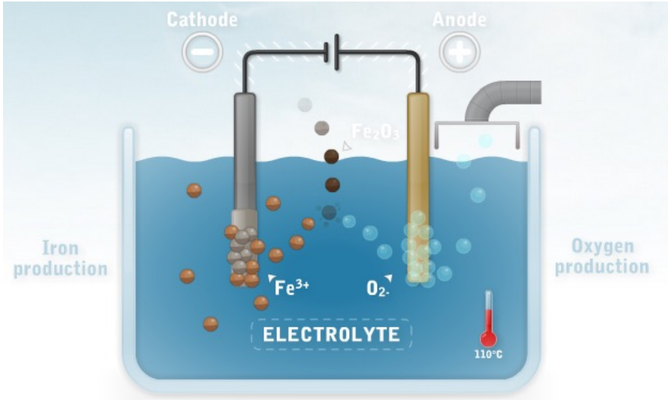


- 40% Secondary route, electric arc furnaces melt scrap metal   0.03 tCO_2/t steel

Possible alternative route in the future:

Direct Reduced Iron (DRI) : Fe_2O_3+2H_2 → 2FeO +2H_2O

FeO + H_2 → Fe + H_2O



Hydrogen-based DRI is being explored in H2Future, HYBRIT, and SALCOS projects.

Smelting, furnaces, refining and rolling, and products finishing can be electrified.

Chemical industry

This sector includes the production of basic organic (olefins, alcohols, aromatics) and inorganic compounds (ammonia, chlorine), polymers (plastics) and end-user products (cosmetics, pharmaceutics).

CO_2 emissions = energy supply + process emissions + carbon embedded in plastics

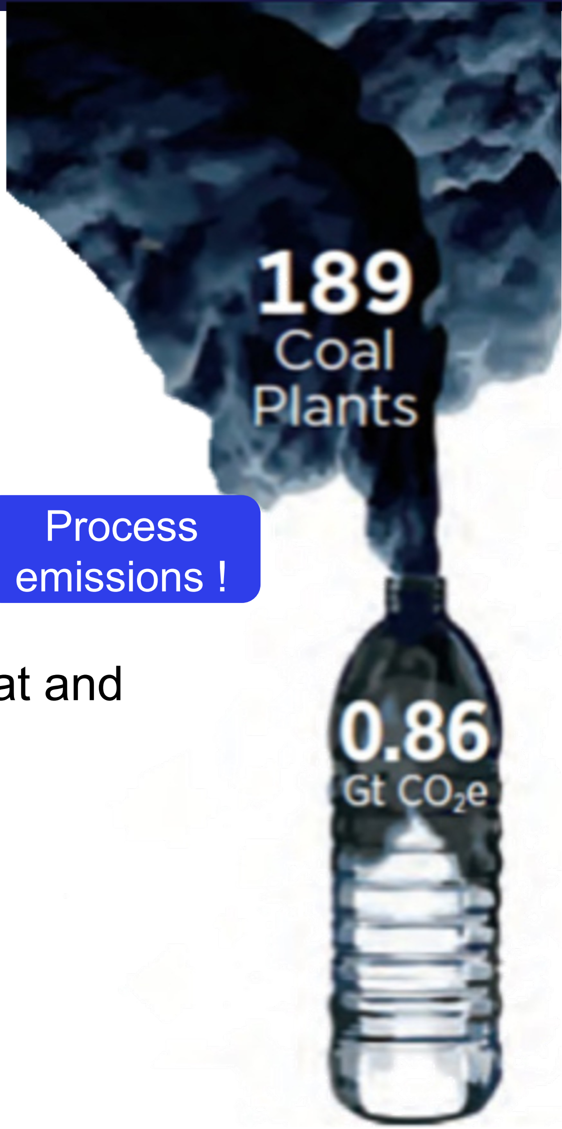
Process emissions !

Energy-related emissions can be reduced using biomass to supply high-temperature heat and electrification of other activities.

Chemical industry consumes large amounts of fossil-fuel based products as feedstock.

Synthetic hydrocarbon feedstock could be produced via Fischer-Tropsch process

$$(2n+1)\, H_2 + n\, CO \rightarrow C_nH_{2n+2} + n\, H_2O$$



2019 plastic emissions in USA, Plastic & Climate

Chemicals industry - Ammonia

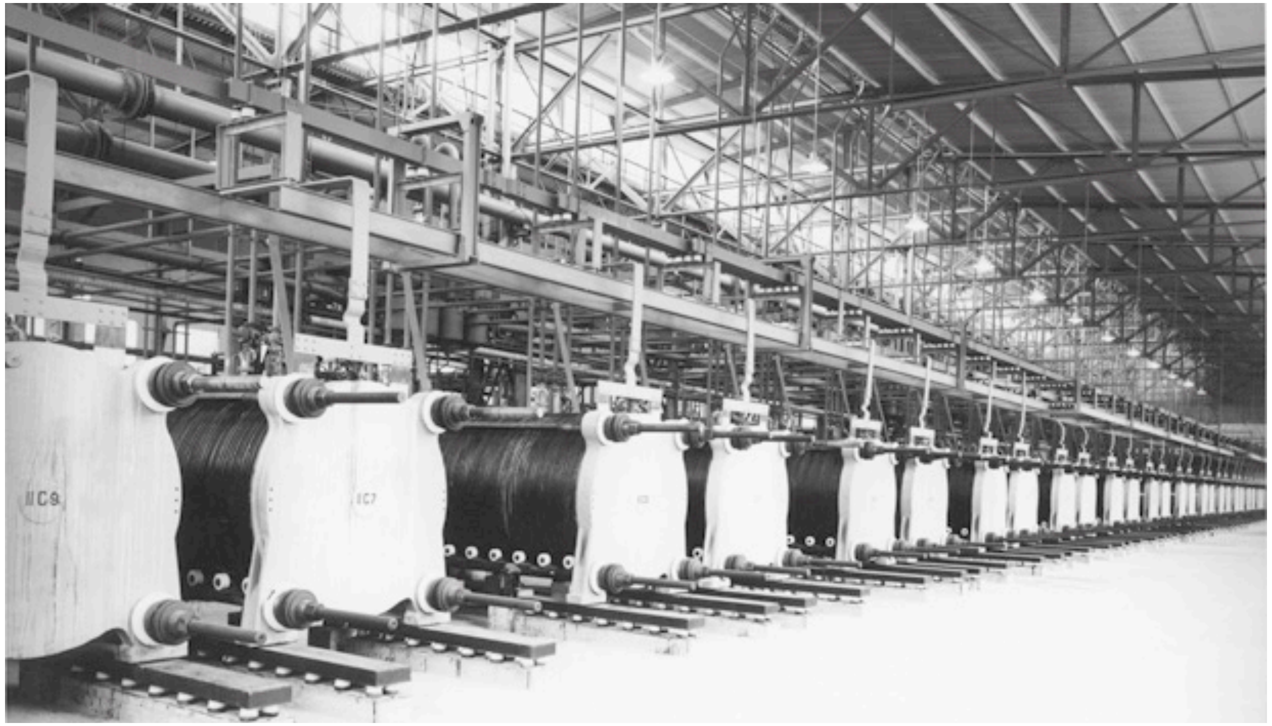
Current situation:

Today almost all the hydrogen is obtained by steam reforming of methane but electrolysers can be used.

Possible alternative route in the future:

Ammonia can be produced via Haber-Bosch process and using green hydrogen

$N_2 + 3\ H_2 \rightarrow 2\ NH_3$



135 MW electrolysers for Ammonia production in Glomfjord, Norway (1953-1991), NEL

Cement, ceramics and glass

Current situation:

Calcination of limestone to lime: $CaCO_3 \rightarrow CaO + CO_2$ 0.54 tCO_2/t cement

Process emissions !

Possible alternative route in the future:

Alternatives at an early development stage: Carbon Capture and Storage (CCS), using new raw materials (wood constructions) or new cement alternatives

For cement, energy-related emissions can be reduced using biomass to supply high-temperature heat and electrification of other activities.

Ceramics and glass manufacturing could be electrified.

Aluminium and other non-ferrous metals

Current situation:

This sector includes the manufacturing of base metals (aluminium, copper, lead or zinc), precious metals (gold, silver), and technology metals (molybdenum, cobalt, silicon).

Aluminium (50% of this sector)

- 40% Primary route, bauxite (aluminium ore) → alumina → aluminium : $2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$

Process emissions !

- 60% Secondary route, scrap metal is re-melted

Possible alternative route in the future:

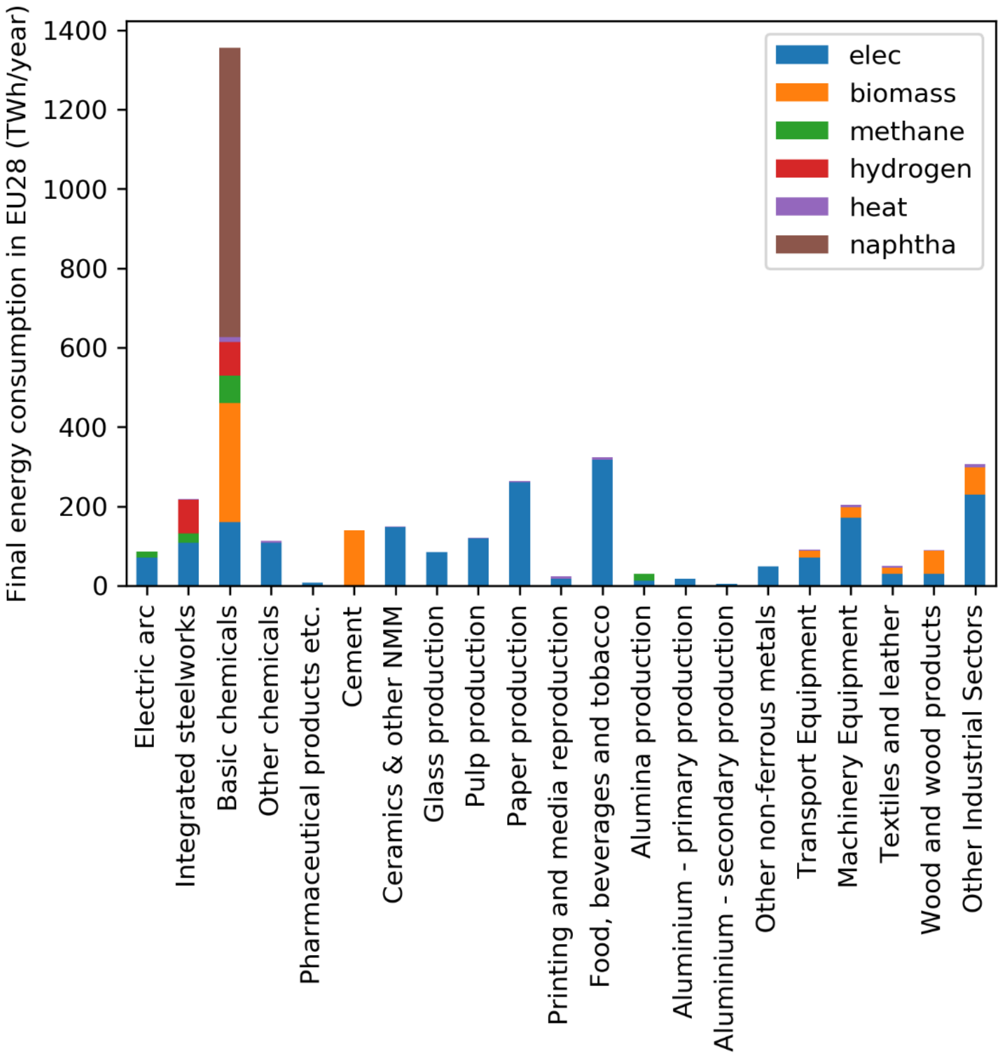
Energy-related emissions can be reduced using methane to supply high-temperature heat and electrification of other activities.

Other industrial sectors

- Pulp, paper and printing
- Food, beverages and tobacco
- Machinery equipment
- Transport equipment
- Wood and wood products
- Textiles and leather
- Other industrial sectors

Most of the activities can be electrified, biomass could also provide process heat.

Future scenario: low-carbon industry consumption



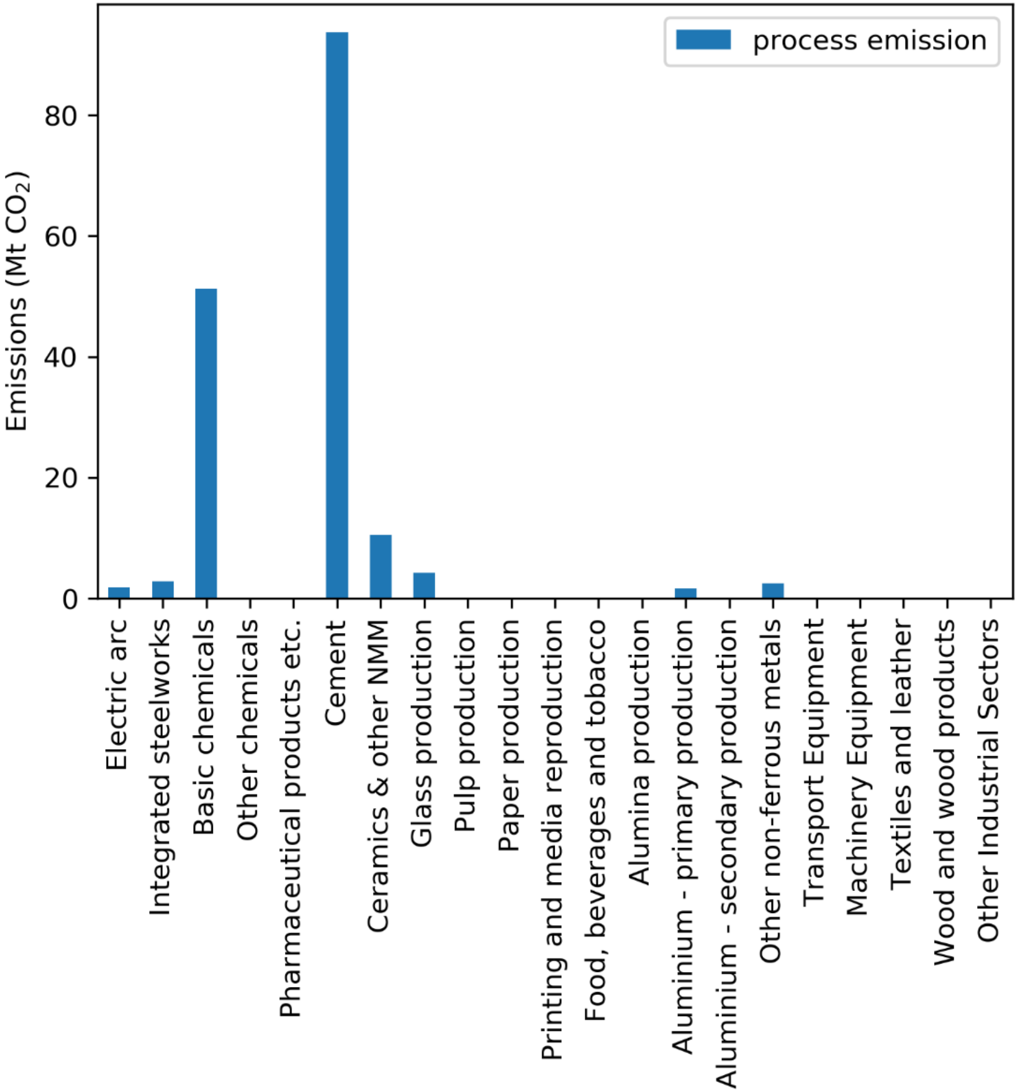
Victoria et al., Joule (2021)

European industry can be decarbonised by

- Electrifying most of the activities, increases electricity consumption by 1027 TWh/y (36% current EU electricity consumption),
- Biomass* (624 TWh/y) and methane (123 TWh/y) for high-temperature processes,
- 58 TWh/y low-temperature heat,
- 728 TWh/y naphtha-equivalent

* Solid biomass sustainable potential =1263 TWh/y (JRC Bioenergy potentials)

Future scenario: low-carbon industry consumption



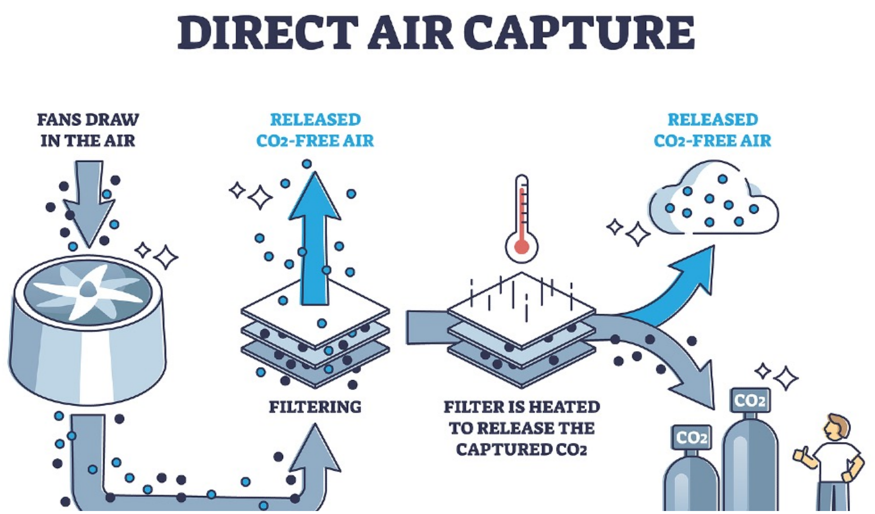
Process emissions still represents 169 Mt CO_2 (20% of total CO_2 emissions from industry in 2015)

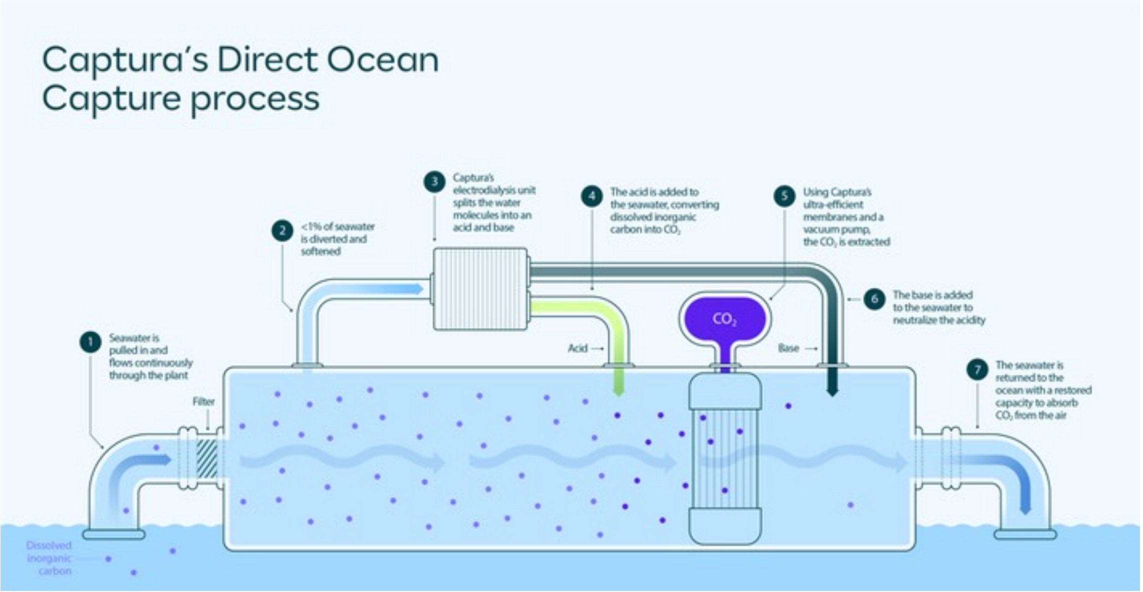
Carbon capture, conversion and storage

CO_2 capture

CO_2 can be sequestered from point-sources: Combined Heat and Power (CHP), or power plants, industry (e.g. cement) .

It can also be captured from diluted sources using technologies like Direct Air Capture (DAC) or Direct Ocean Capture (DOC).



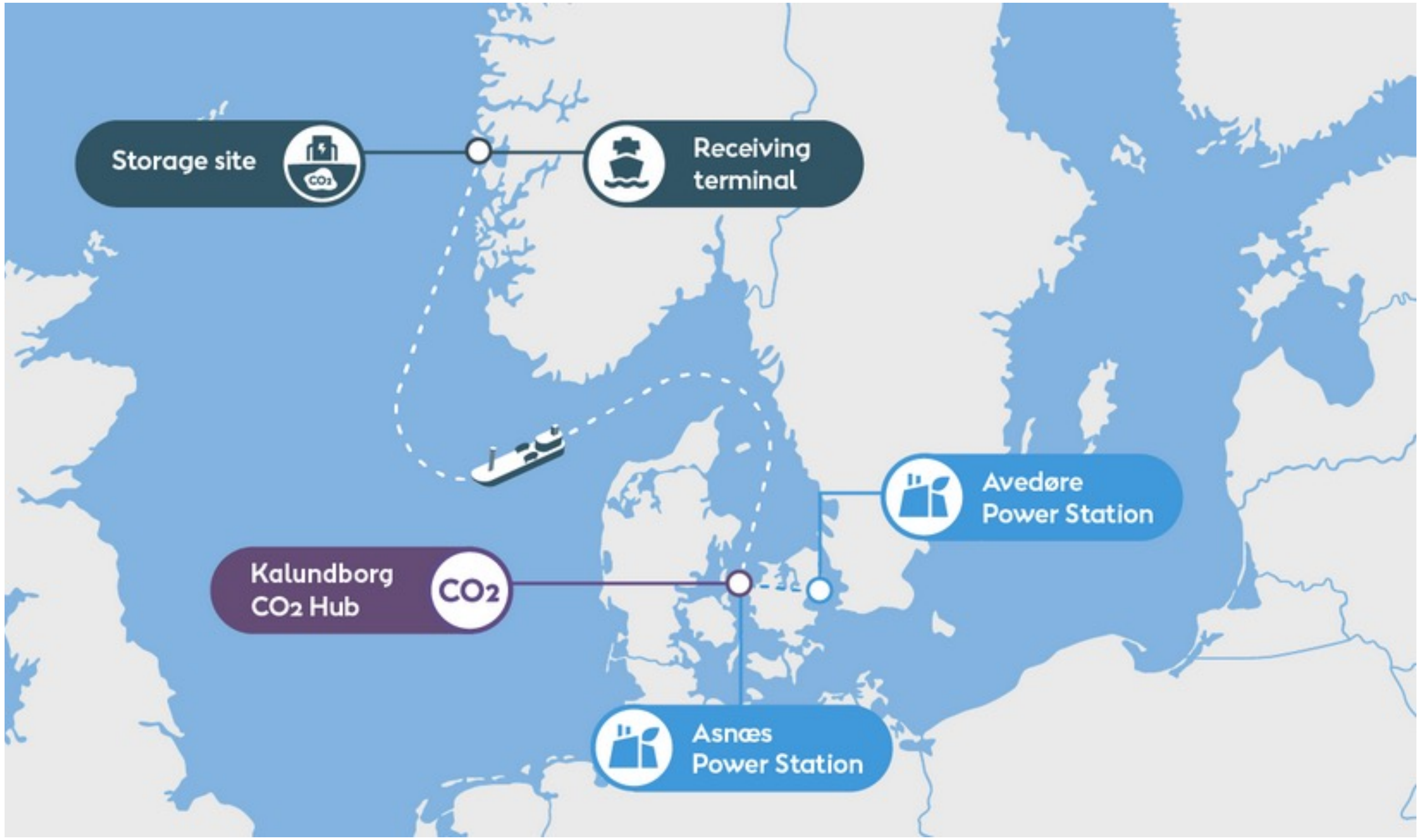


CO_2 conversion or sequestration

Captured CO_2 can be used to produced synthetic fuels (methane, methanol, oil).

Or it can be sequestered underground.

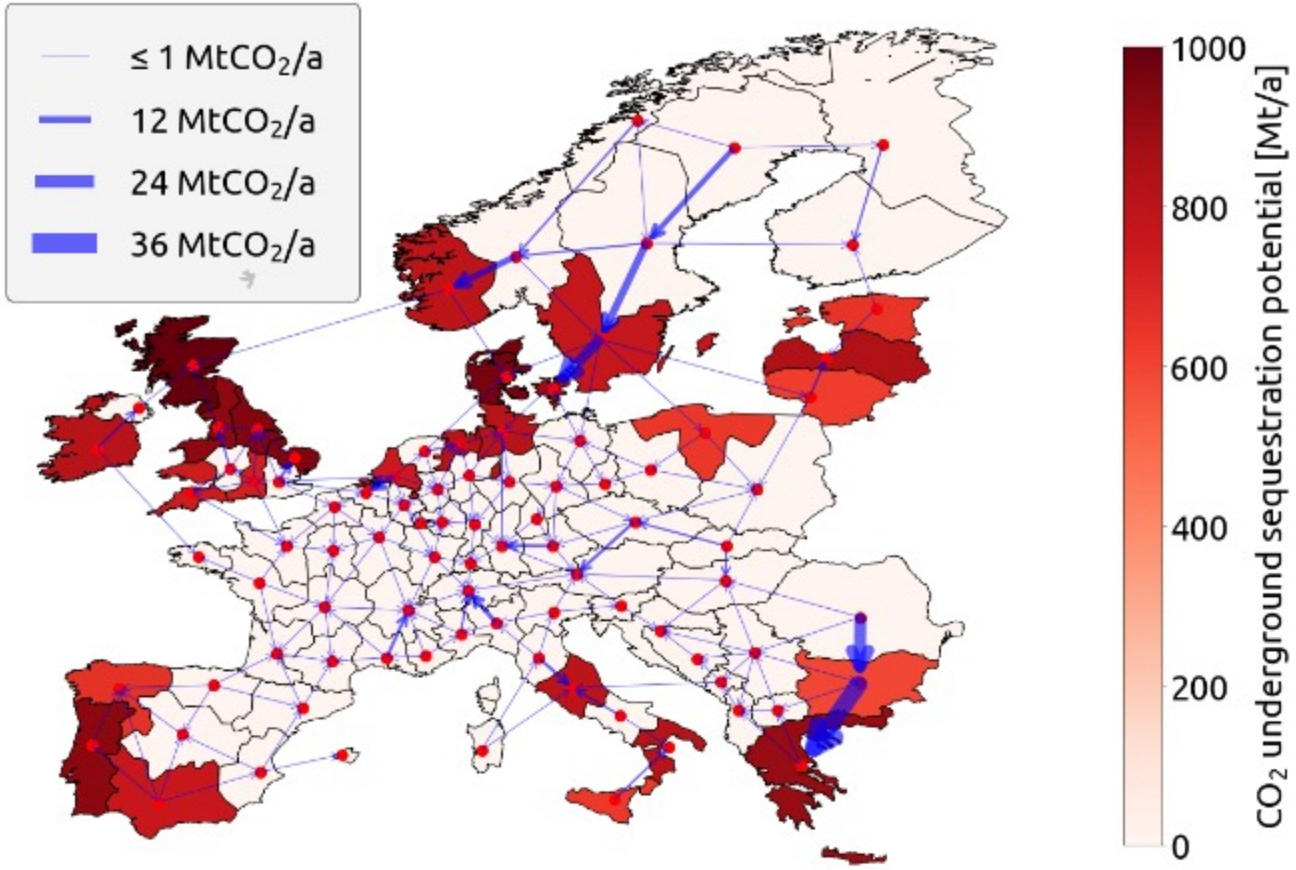
'Ørsted Kalundborg CO2 Hub' :
From the summer* of 2026, 430,000 tonnes of CO_2 will be captured annually from two heat and power plants and store it in the North Sea.



* Predicted to start in the spring/summer of 2026 at full operations.

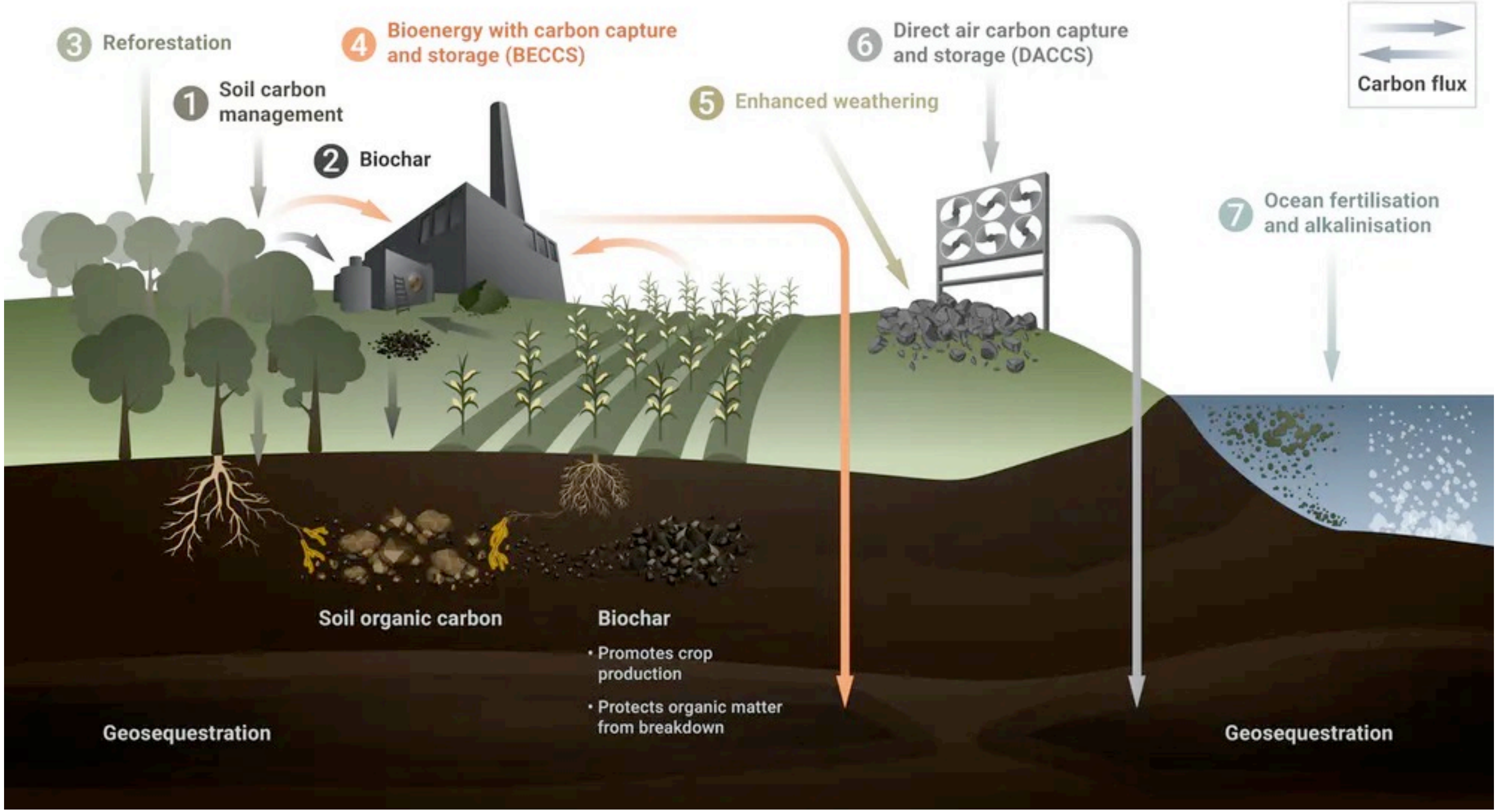
Should we transport CO_2 across Europe?

It can be cost-effective to transport CO_2 using a network of pipelines build for that purpose.



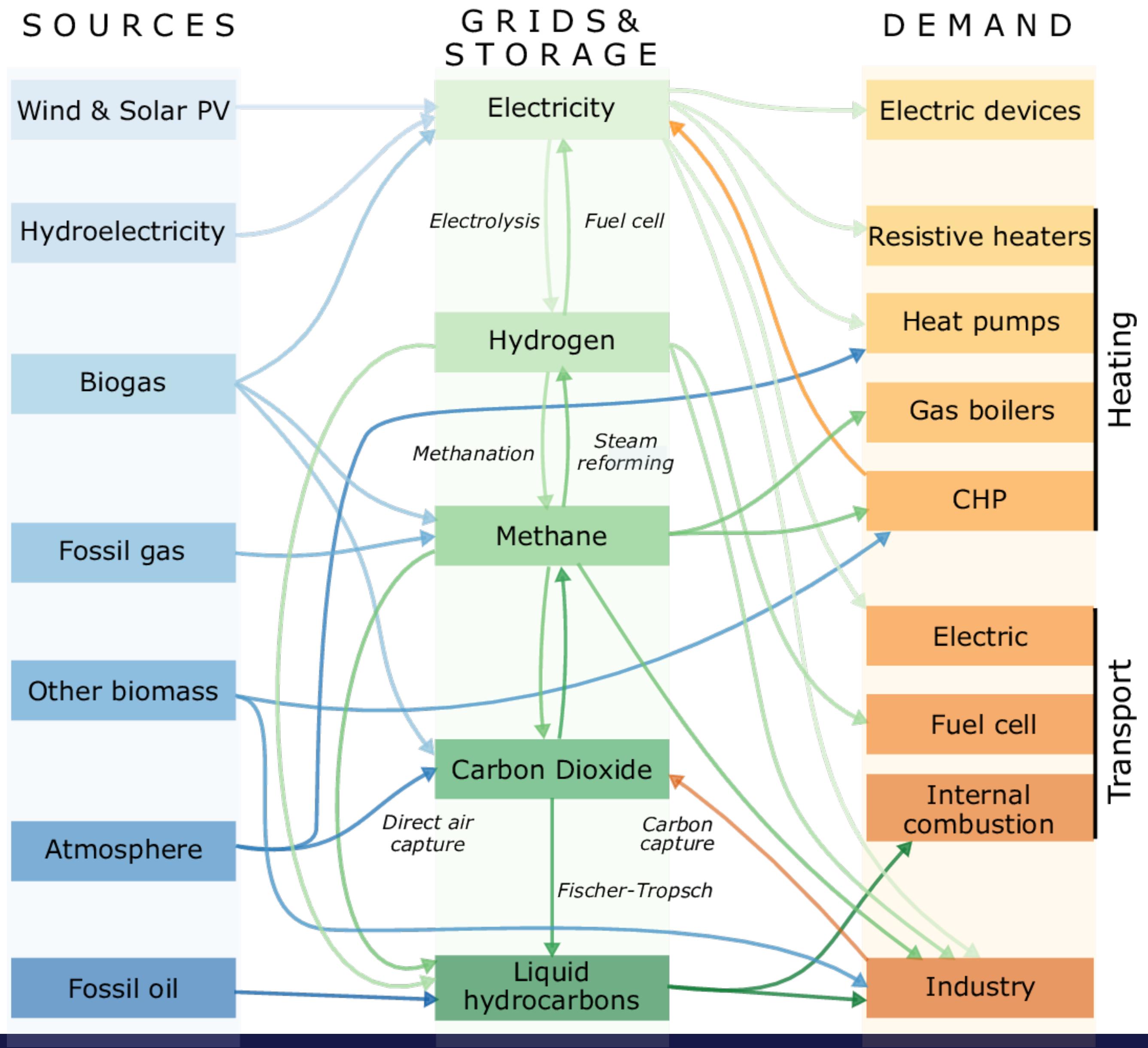
Fernandes et al (under review), 2024, ArXiv

Other CCU strategies outside the energy sector



Ref: The conversation

Energy and carbon balancing



Problems for this lecture

Problems 12.1 (**Group 25**)

Problems 12.2 (**Group 26**)

Next week: Some current research topics + suggestions for MSc theses

NB: the course evaluation is open, we care about your feedback, and we'd appreciate feedback on the practical part of the course + usage of AI

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